

eFrei

PARIS PANTHÉON-ASSAS UNIVERSITÉ

Probabilities Project

Mathematics Department

(in coordination with the Informatics Department)

2026/2027 S3 - I2-New

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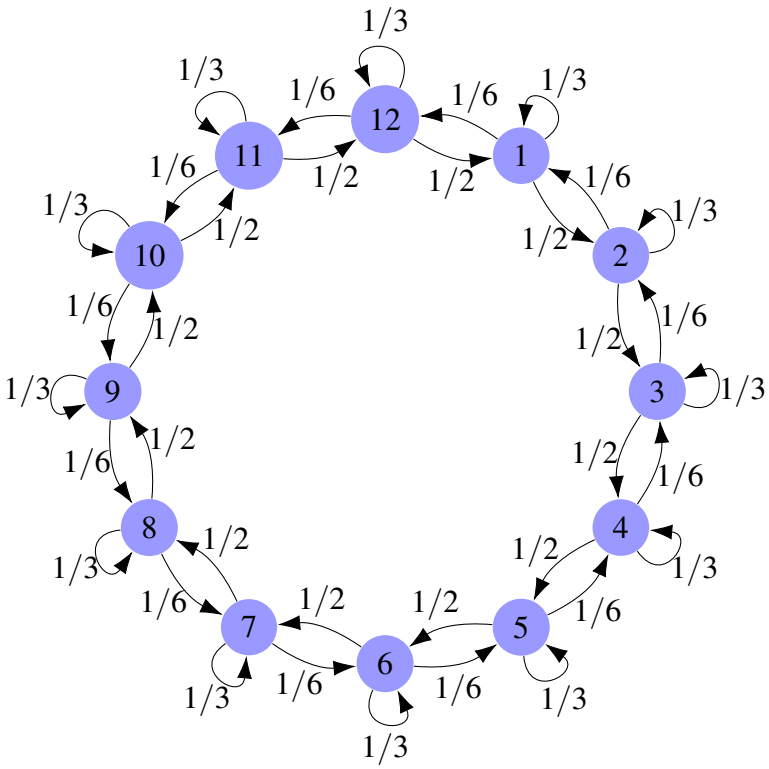
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1. Introduction to Markov chains

1.1 Introductory examples

Consider a broken clock, with a single hand, that each hour can either move forward one hour with probability $1/2$, stay in place with probability $1/3$, or move back one hour with probability $1/6$. The evolution in time of the clock is a random process which can be represented as a graph:

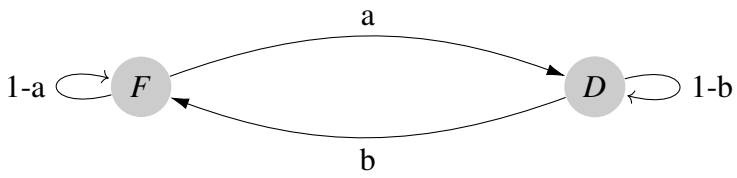


The exact same information can be represented in the form of a matrix, called the *transition matrix* of the system or the *adjacency matrix* of the graph:

$$\begin{pmatrix}
\frac{1}{3} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{6} \\
\frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{6} & \frac{1}{3} & \frac{1}{2} \\
\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{6} & \frac{1}{3}
\end{pmatrix}$$

The coefficient in the i-th row and j-th column represents the weight of the arrow from i to j.

A simpler example of a Markov chain was hidden in exercise 2.23 of the lecture notes: at instant $n\Delta t$, the machine can be either functional (F_n) or damaged (D_n) and the transition probabilities are given by the following graph



This is called the two-states Markov chain. The probability that the machine is functional at instant $n\Delta t$ is

$$\begin{aligned} P(F_n) &= P(F_n \,|\, F_{n-1})P(F_{n-1}) + P(F_n \,|\, D_{n-1})P(D_{n-1}) \\ &= (1-a)P(F_{n-1}) + bP(D_{n-1}) \end{aligned}$$

Similarly, the probability that the machine is damaged at instant $n\Delta t$ is

$$P(D_n) = aP(F_{n-1}) + (1-b)P(D_{n-1}).$$

These two equations can be written together using row vectors and a square matrix:

$$\begin{pmatrix} P(F_n) & P(D_n) \end{pmatrix} = \begin{pmatrix} P(F_{n-1}) & P(D_{n-1}) \end{pmatrix} \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}.$$

1.2 The framework of Markov chains

In opposition to a deterministic system, a *stochastic process* is used to describe the dynamics of a system (the clock's hand, the machine) that exhibits some sort of *random behaviour*: given that the system is in the state s_i at time t_0, one can at best determine the probabilities $Pr(X_{t_1} = s_f \,|\, X_{t_0} = s_i)$ that the system will be found in state s_f at a later time t_1. Finite Markov chains are stochastic processes satisfying the following three properties:

1) **Discreteness of time:** time is described as a parameter t taking values in some subset of $\mathbb{N}$.
2) **Finite space of states:** the set of all states accessible to the system, called the *space of states* and denoted by S, is finite.
3) **Markov property:** the probability for the system to be in a certain state at time $t+1$ depends only on the state of the system at time t. In other words, only the information about the present state of the system is necessary to predict the future behaviour of the system.

In addition, a time-homogeneous finite Markov chain satisfies the following fourth property:

4) **Time homogeneity:** the transition probabilities to go from one state to another in one step are independent of time. In other words, for any $t_1, t_2 \in \mathbb{N}^*$, we have

$$Pr(X_{t_1} = j \,|\, X_{t_1-1} = k) = Pr(X_{t_2} = j \,|\, X_{t_2-1} = k).$$

If we denote by n the cardinality of the space of states S ($n = 12$ in the first example above), the fundamental equation that captures the dynamics of a Markov chain is

$$Pr(X_t = j) = \sum_{k=1}^{n} Pr(X_t = j \,|\, X_{t-1} = k) Pr(X_{t-1} = k)$$

(this is simply the total probabilities formula). This equation can be written much more elegantly using the tools of linear algebra : if we define

- the *row vector* $\Pi(t)$, called *state distribution*, whose j-th component is $Pr(X_t = j)$,
- the $n \times n$ *transition matrix* P whose coefficient of row k and column j is $Pr(X_t = j \,|\, X_{t-1} = k)$,

then, the fundamental law for Markov chains becomes

$$\boxed{\Pi(t+1) = \Pi(t)P}.$$

Remarks: As immediate consequences of the definitions, we see that

⋆ P is a positive matrix such that the sum of all coefficients of any row is equal to 1:

$$\sum_{j=1}^{n} P_{jk} = \sum_{j=1}^{n} Pr(X_1 = j \,|\, X_0 = k) = 1.$$

A matrix satisfying these two properties is called a *right stochastic matrix*.

⋆ For any $l \in \mathbb{N}^*$, the probability distribution l steps after the present state is determined by

$$\Pi(t+l) = \Pi(t)P^l.$$

In other words, the coefficient $(P^l)_{kj}$ denotes the probability for the system to be in state k l steps after being in state j.

1.3 Real-life examples

Examples of dynamical processes that can be modelled using Markov chains are found in almost all areas:

- The weather: states are "sunny", "cloudy", "rainy", etc, and the time-step can be every ten minutes.
- The long-term evolution of an ecosystem: states are the different vegetations found in a given area (for example, oak grove, vineyards, grass, Pine forest) and the time-step can be every ten years.
- The position of a Velib in Paris: states are the different Velib stations in the city and the time-step can be every 30 minutes.
- Simple dice games as snakes-and-ladders: states are the different squares where the piece can be located and the time-step is every dice throw.
- Stock prices in Wall-street: states are the different possible values for a stock and the time-step is every second.
- Brownian motion in a closed lattice (random walk of a particle): state are the different possible positions of the particle and the time-step might be every micro-second.

1.4 Goal of the project

In this project, we want you to explore the long-run behaviour of Markov chains. More precisely, given a Markov chain with transition matrix P, and given an initial state distribution $\Pi(0)$, under which conditions does the limiting distribution $\lim_{t\to+\infty} \Pi(t) = \lim_{t\to+\infty} \Pi(0)P^t$ exist?

2. Numerical exploration

We will start by exploring "experimentally" what is the long-term behaviour of a particular (and seemingly complicated) Markov chain. Here is the 27-states transition matrix we will work with:

$$\begin{pmatrix}
0 & 0 & 0 & 0.6 & 0 & 0 & 0 & 0 & 0 & 0 & 0.2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.2 \\
0 & 0.3 & 0 & 0 & 0.2 & 0 & 0 & 0 & 0 & 0 & 0 & 0.4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0.4 & 0.3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.3 & 0 & 0 & 0 & 0 \\
0.6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.2 & 0 \\
0 & 0.4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.2 & 0 & 0 & 0.3 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0.3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.4 & 0 & 0 & 0.3 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.1 & 0 & 0 & 0 & 0 & 0 & 0 & 0.9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.4 & 0 & 0 & 0 & 0 & 0 & 0 & 0.2 & 0 & 0 & 0.2 & 0 & 0.2 & 0 & 0 & 0 \\
0.5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0.1 & 0 & 0 & 0.4 & 0 & 0 & 0 & 0 & 0 & 0 & 0.3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.2 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0.3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.7 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0.1 & 0 & 0 & 0.2 & 0 & 0 & 0.1 & 0.1 & 0 & 0 & 0 & 0 & 0.4 & 0 & 0.1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.5 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0.5 & 0 & 0 & 0 & 0 & 0 & 0 & 0.5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.5 & 0 & 0 & 0.1 & 0 & 0.3 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0.2 & 0 & 0 & 0.3 & 0 & 0 & 0 & 0 & 0 & 0 & 0.1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.1 & 0 & 0 & 0 & 0.3 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.2 & 0 & 0 & 0 & 0 & 0 & 0.4 & 0 & 0 & 0.2 & 0 & 0 & 0 & 0 & 0.2 & 0 & 0 & 0 \\
0 & 0 & 0.3 & 0 & 0 & 0 & 0.3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.4 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.3 & 0 & 0 & 0 & 0.1 & 0 & 0 & 0 & 0 & 0.3 & 0 & 0 & 0.3 & 0 & 0 & 0 & 0 & 0 \\
0 & 0.2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.5 & 0 & 0 & 0 & 0.1 & 0 & 0 \\
0 & 0 & 0 & 0.5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.5 \\
0.5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.5 & 0 & 0
\end{pmatrix}$$

(You can find on Moodle a `.txt` file with the matrix given to you in the exact same format you use in the project of the *Data Structure and Programming 2* course.)

Question 1

Suppose that the system starts in state 2.

a) Use the `matrix.c/matrix.h` files you developed in the *Data Structure and Programming 2* course to determine the state distribution after $n = 1$, $n = 2$, $n = 10$, $n = 50$ steps[1].

b) Plot the state distribution $\Pi_A(n)$ as a function of n.

c) Determine whether a limiting distribution exists. If it does, provide the values of the limiting distribution

Question 2

Suppose now that the system starts evenly distributed between the states 2, 5, 12, 21 and 25 (that is, $P(X_0 = 2) = P(X_0 = 5) = P(X_0 = 12) = P(X_0 = 21) = P(X_0 = 25) = \frac{1}{5}$). Does a limiting distribution exist?

Question 3

Suppose now that the system starts randomly distributed between the states 2, 5, 12, 21 and 25 (that is, $P(X_0 = 2) = a$, $P(X_0 = 5) = b$, $P(X_0 = 12) = c$, $P(X_0 = 21) = d$, $P(X_0 = 25) = e$ and $a+b+c+d+e = 1$). Analyse whether a limiting distribution exists and, if so, determine how it depends on the initial–state parameters a, b, c, d and e. Questions 1 to 3 converge

Question 4

Same questions if the system starts in state 8, evenly distributed between states 8, 9 and 16, or randomly distributed between states 8, 9 and 16. All converge

Question 5

Same questions if the system starts in state 14, evenly distributed between states 10, 14, 19, 22 and 24, or randomly distributed between these states. All converge

Question 6

Same questions if the system starts in state 6, evenly distributed between states 6, 17 and 20, or randomly distributed between these states. Anyone converges

Question 7

Same questions if the system starts in state 3, evenly distributed between states 3, 7 and 23, or randomly distributed between these states. State 3 and evenly distributed converges // randomly distributed does not

[1] If you have not yet finished part 3 of the Data Structure and Programming 2 project, or if your own program does not work, then use Matlab and its documentation for the computations (Help for matrices and Help for plots).

3. Making sense of the results

In the previous section, you should have observed that sometimes the Markov chain admits a limiting distribution and sometimes it does not, sometimes the limiting distribution depends on the values of the parameters and sometimes it does not. In this section, we will introduce the theoretical concepts that will help you structuring this results.

3.1 The key concepts

• **Accessible states**: We say that state j is accessible from state k if, starting from state k, it is possible to reach state j after a certain number of steps:

$$\exists n \in \mathbb{N},\ Pr(X_n = j \,|\, X_0 = k) = (P^n)_{\substack{kn \\ kj}} \neq 0.$$

Notice that, since $(P^0)_{kk} = 1$, any state is accessible from itself no matter what.

• **Communicating states**: We say that states j and k communicate if j is accessible from k and vice-versa.

• **Communicating classes**: communicating is an equivalence relation that breaks the space of states into equivalence classes called the communicating classes.

• **Successive classes**: Let C_1 and C_2 be two different communicating classes of a same Markov chain. We say that C_2 is a successor of C_1 if there exists states $k \in C_1$ and $l \in C_2$ such that

$$Pr(X_1 = l \,|\, X_0 = k) = P_{kl} \neq 0.$$

• **Persistent and transient classes**: a communicating class is said to be persistent if it has no successors. Otherwise, it is called a transient class. Once the system reaches a persistent class, it cannot leave it.

• **Period of a state/class**: Let $k \in S$ be a state and consider the set N_{kk} defined as

$$N_{kk} = \left\{ n \in \mathbb{N}^* \,\middle|\, (P^n)_{kk} \neq 0 \right\}.$$

In other words, a natural number n belongs to N_{kk} if it is possible to *return* to state k in n steps. Then, the *period* of the state k is defined as the greatest common divisor of all such numbers:

$$d_k := \gcd(N_{kk}).$$

If $d_k = 1$, the state is said to be *aperiodic*.

Two states belonging to the same communicating class have the same period. Therefore, one can talk about the period of a communicating class.

• **(Ir)reducible Markov chains**: a Markov chain is said to be irreducible if it contains one single communicating class and reducible otherwise.

• **Reduced graph of a reducible Markov chain**: given a reducible Markov chain, its reduced graph (or *Hasse diagram*) is the graph of the successor relation on communicating classes. In other words, its vertices are the communicating classes and there exists an edge from C_k to C_l if and only if C_l is a successor of C_k.

• **Stationary distribution**: Given a finite Markov chain with transition matrix P, a probability distribution is said to be stationary if it satisfies

$$\Pi = \Pi P.$$

(In other words, (the transposed of) Π is an eigenvector of (the transposed of) P associated to the eigenvalue 1.)

• **Limiting distribution**: Given a finite Markov chain with transition matrix P, a probability distribution Π is said to be a limiting distribution if there exists an initial distribution $\Pi(0)$ such that

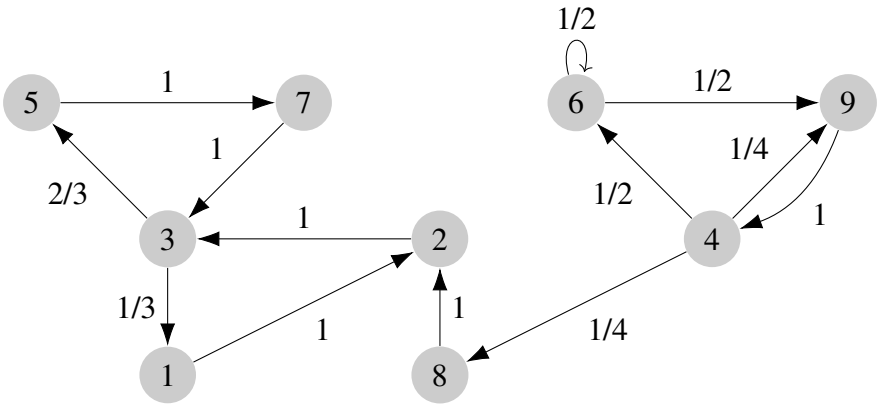
$$\Pi = \lim_{n \to +\infty} \Pi_0 P^n.$$

EXAMPLE:

Consider the Markov chain defined by the following transition matrix

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1/3 & 0 & 0 & 0 & 2/3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 & 0 & 1/4 & 1/4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 & 0 & 0 & 1/2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Notice that for any given row the sum of all coefficients is indeed equal to 1. The graph associated to this Markov chain is



The chain is reducible since it contains three communicating classes: $\{1,2,3,5,7\}$, $\{4,6,9\}$ and $\{8\}$. $\{8\}$ is the only successor of $\{4,6,9\}$ whereas $\{1,2,3,5,7\}$ is the only successor of $\{8\}$. Therefore, the communicating classes $\{4,6,9\}$ and $\{8\}$ are transient and the class $\{1,2,3,5,7\}$ is persistent. The reduced graph of this Markov chain is:

$$\{1,2,3,5,7\} \longleftarrow \{8\} \longleftarrow \{4,6,9\}.$$

We focus now on the class $\{4,6,9\}$. We have

$$\begin{aligned} N_{44} &= \{2,3,4,5,6,\ldots\}, \\ N_{66} &= \{1,2,3,4,5,6,\ldots\}, \\ N_{99} &= \{2,3,4,5,6,\ldots\}. \end{aligned}$$

Therefore, this class is aperiodic. On the other hand, for the class $\{1,2,3,5,7\}$, we have:

$$N_{11} = N_{22} = N_{33} = N_{55} = N_{77} = \{3,6,9,12,\ldots\}.$$

Therefore, this class has a period of 3.

3.2 Back to the 27-states Markov chain

Question 8

Apply the programs developed in part 2 of the Data Structure and Programming project to the 27-states Markov chain given in the previous section in order to:

a) Identify the different communicating classes.

b) Draw the reduced graph (Hasse diagram) of the whole Markov chain.

c) For each communicating class, determine if it is transient or persistent, draw its graph and compute its period.

Question 9

Interpret the answers to questions 1 through 7 in the light of this new perspective.

Question 10

Only one persistent class was not explored in questions 1 through 7. Find the set of all stationary distributions of that communicating class.

4. To go further

4.1 Limiting distributions starting from transient states

Question 11

Denote respectively by $L\Pi_A$, $L\Pi_B$ and $L\Pi_C$ the limiting distributions obtained when the system starts at state 2, state 8 or state 14. Prove that $L\Pi_C$ is a linear combination of the other two limiting distributions:

$$\exists(\alpha,\beta)\in\mathbb{R}^2, L\Pi_C = \alpha L\Pi_A + \beta L\Pi_B.$$

Can you give a probabilistic interpretation to the coefficients α and β ?
Can you compute by hand (without a computer!) the exact values of α and β?

Question 12

Let s be a state and C_k an absorging communicating class. Denote by $s \to C_k$ the event: "the system starts from state s and reaches the class C_k". For each state 10, 19, 22, 24, estimate by hand the values of $P(s \to C_k)$ (for all absorbing classes), then find the "exact" value using the computer.

4.2 Some proofs

Question 13 – *The period of a communicating class*

Prove that two states belonging to the same communicating class must have the same period.

Question 14 – *The number of limiting distributions*

Consider a Markov chain with n states. Let $\mathscr{L} \subset \mathbb{R}^n$ and $\mathscr{S} \subset \mathbb{R}^n$ denote respectively the sets of all limiting distributions and stationary distributions.

a) Prove that $\mathscr{L} = \mathscr{S}$.

b) A subset E of $\mathbb{R}^n$ is convex if

$$\forall (u,v) \in E, \forall \alpha \in [0,1],\ \alpha u + (1-\alpha)v \in E.$$

Prove that $\mathscr{L}$ is convex.

c) Using your course of linear algebra of the last semester (but no more sophisticated tools!), prove that a finite irreducible Markov chain admits at least one stationary distribution.

d) Use the previous two questions to prove that a reducible Markov chain admits infinitely many limiting distributions.

e) Prove that a d-periodic irreducible Markov chain (with $d \neq 1$) does not always admit a limiting distribution.

f) Look up for a proof that, for an aperiodic irreducible Markov chain, a limiting distribution always exists and is unique, and see if you can understand all the details of the proof!

5. Deliverable

The project is carried out in teams of three to five students, with at most eight teams per TD group. As already requested in class, students must form their teams and register them in the dedicated section on Moodle.

5.1 Report

Each team is required to submit a written report of no more than 20 pages, answering the questions of the previous sections. Grading will take into account presentation (use of LaTeX is encouraged), the quality of the plots presented, the conciseness of answers and interpretations.

Each team must email its report directly to the instructor by Saturday, October 31, 2026 at 23:59 (Paris time). Do not submit on Moodle. This is a firm deadline; late submissions will be heavily penalised.

5.2 The oral presentation

The oral presentations will take place on Monday, November 2, 2026.

The oral presentation will be a 10-minute lecture on the existence of limiting distributions in Markov chains. It should not be a repetition of what is written in the report but rather a synthesis of everything you understood working in this project. You should provide new examples and run your programs to illustrate your lecture.

Slides are expected for this lecture. Of course, slides must not include any handwritten or photographed text. All formulae and matrices must be typed. Pedagogy and clarity will be be valued.

At the end of the presentation, there will be 15 minutes for questions: each student can be questioned individually on the project or on a point from the course.